Hyperspectral Image Classification Based on Multibranch Attention Transformer Networks

Jing Bai, *Senior Member, IEEE*, Zheng Wen, Zhu Xiao, *Senior Member, IEEE*, Fawang Ye, Yongdong Zhu, Mamoun Alazab, *Senior Member, IEEE*, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— Deep learning (DL) has become a mainstream method of hyperspectral image (HSI) classification. Many DL-based methods exploit spatial-spectral features to achieve better classification results. However, due to the complex backgrounds in HSIs, existing methods usually show unsatisfactory performance for the class pixels located on the land-cover category boundary area. In large part, this is because the network is susceptible to interference by the irrelevant information around the target pixel in the training stage, resulting in inaccurate feature extraction. In this article, a new multibranch transformer architecture (spectral spatial transformer (SST)-M) that assembles spatial attention and extracts spectral features is proposed to address this problem. The transformer model has a global receptive field and thus can integrate global spatial position information in the HSI cube. Meanwhile, we design a spatial sequence attention model to enhance the useful spatial location features and weaken invalid information. Considering that HSIs contain considerable spectral information, a spectral feature extraction model is designed to extract discriminative spectral features, replacing the widely used principal component analysis (PCA) method and obtaining better classification results than it. Finally, inspired by semantic segmentation, a mask prediction model is designed to classify all of the pixels in the HSI cube; this guides the neural network to learn precise pixel characteristics and spatial distributions. To verify the effectiveness of our algorithm (SST-M), quantitative experiments were conducted in three well-known datasets, namely, Indian Pines (IP), University of Pavia (PU), and Kennedy Space Center (KSC). The experimental results demonstrate that the proposed model achieves better performance than the other state-of-the-art methods.**

***Index Terms*— Deep learning (DL), hyperspectral image classification (HSIC), multibranch prediction, self-attention (SA) mechanism, spatial attention, transformer model.**

I. Introduction

HYPERSPECTRAL image (HSI) classification is an important research subject in the remote sensing field. The rich spectral information contained in HSIs can compensate for the deficiency of the spectral resolution in optical images and conventional remote sensing data; thus, hyperspectral imaging is clearly advantageous for many remote sensing applications, such as early warning of forest disasters, geomorphic transformation detection, and urban monitoring [1], [2].

The aim of hyperspectral image classification (HSIC) is to distinguish the category of each pixel in HSIs. Early works either concentrate on statistical analysis [3] or leverage simple pattern recognition algorithms such as k-nearest neighbors (K-NNs) [4], random forests (RFs) [5], and support vector machines (SVMs) [6]. Then, some researchers have devised Markov random field (MRF)-based methods [7] and morphological profile-based methods [8], [9] to exploit the spatial location relationship in HSIs. Since 2014, with the remote sensing data analysis moved into the deep learning (DL) era, lots of DL-based methods for HSI classification have gradually become mainstream solutions. The DL approaches applied HSI into three aspects [10]: convolutional-based, such as 2-D convolutional neural network (CNN) [11], 3-D CNN [12]; generative-based, such as generative adversarial networks (GANs) [13]; sequence-based, such as recurrent neural networks (RNNs) [14], and long short term memory network (LSTM). Subsequently, some deeper networks and more complex models have also been designed; for example, Huang *et al*. [15] present a two-phase network constructed by two DenseNets. He *et al*. [16] use deep pyramidal residual networks to improve the spectral-spatial features in HSIs. A number of novel methods, including graph convolution networks (GCNs) have also been designed for HSI classification. In [17], the authors propose a multisource compensation network (MSCN) to tackle distribution discrepancies and category incompleteness that alleviates domain shift issues by mapping the two domain features into a common feature space. Multistructure Unified Discriminative Embedding (MUDE) [19]

Manuscript received 16 May 2022; revised 16 June 2022; accepted 27 July 2022. Date of publication 5 August 2022; date of current version 15 August 2022. This work was supported in part by the Key Research and Development Program of Shaanxi under Grant S2022-YF-YBGY-0921 and Grant 2020GXLH-Y-023, in part by the National Natural Science Foundation of China under Grant 61772401, in part by the Key Research and Development Project of Hunan Province of China under Grant 2022GK2020, in part by the Funding Project of Zhejiang Laboratory under Grant 2021LC0AB05, in part by the Hunan Natural Science Foundation under Grant 2022JJ30171, in part by the Science and Technology Project of Hunan Provincial Water Resources Department under Grant XSKJ2021000-39, in part by the Project of State Key Laboratory (SKL) of Remote Sensing Information and Image Analysis Technology under Grant 6142A010409, and in part by the Scientific Research Project of Hunan Provincial Natural Resources Department under Grant 2022-11. *(Corresponding authors: Zhu Xiao; Yongdong Zhu.)*

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Digital Object Identifier 10.1109/TGRS.2022.3196661

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design the intraclass and interclass neighborhood structure graphs to represent the intrinsic properties of HSI for the complex imaging condition. In recently, considering the problem of limited data, semisupervised-based and incremental learning-based methods are also used in HSIC. In [18], the author develops a new semisupervised feature extract algorithm called a geodesic-based sparse manifold hypergraph (GSMH), which combines the sparse representation, graph embedding, and semisupervised method to depicts the multivariate relationship of samples and tackle the small size sample problem. In [20], the authors propose a novel graph-based semisupervised network; the whole image is used as the input. Several graph convolution layers are used to extract features, which adopts semisupervised learning and can be trained end to end. In [21], a linear programming incremental learning classifier (LPILC) is proposed, which utilize the existing DL classification models to learn new classifying capability rapidly within a few new samples.

DL-based methods classify pixels in HSIs one by one and crop object pixel-centric 3-D pixel cubes as training samples to introduce spatial context information. The category of the center pixel is used as the label of the HSI cube to train the neural network. This is based on the assumption that the center pixel shares the same data characteristics as the pixels within the neighborhood. Thus, most pixels can be accurately classified. However, this approach gives rise to two problems.

1) When the center pixel is located on the category boundary area, an HSI cube often contains many irrelevant pixels that may have different land-cover categories from the center pixel. This leads to a negative impact on the neural network to accurately classify the center pixels.

2) In a land-category crossing area, an HSI cube usually includes multiple pixel categories, and classifying the category of the center pixel may not be sufficient to fully represent the characteristics of the HSI cube. Accordingly, this will cause the neural network to learn inadequate fitting parameters. To address these problems, the vision attention mechanism was integrated with CNN [22], [23] to distinguish useful and irrelevant features adaptively and achieve anticipated advances. CNN-based methods use a small convolution kernel and thus have a limited receptive field because a convolution operation cannot combine all of the spatial information from the HSI cube. The use of more abundant convolution kernels and deeper layers is necessary to acquire long-distance dependency, which will both certainly increase the computational overhead. GCN-based methods [24] optimize the convolution operators to find more appropriate features in spatial locations. However, GCN requires a non-Euclidean structure and a long time for training [25], resulting in additional efforts needed to construct a suitable data structure from the HSI cube. In addition to these methods, some studies apply superpixel segmentation algorithms to HSI cubes so that pixels with the same land-cover category are gathered into superpixel patches [26], [27], thereby avoiding disturbance from irrelevant pixels. In [28], the authors proposed the selection of diverse shapes of pixel patches as inputs to learn diverse contextual features. However, it is difficult to search for a proper superpixel segmentation algorithm or patch shape for different data types, and many hyperparameters rely on manual adjustments. Alternative approaches based on instance segmentation have been implemented to simultaneously distinguish all of the pixels in HSIs. In [29], the DeepLab V2 module is employed to learn abstract semantic segmentation images. Lee and Kwon [30] created fully convolutional networks in which three CNN layers and diverse size kernels were utilized to capture richer contextual information. Sun *et al.* [31] focused on the spatial distribution of all pixels and designed a fully convolutional segmentation network to predict all land-cover categories in the HSI cube.

Although extensive DL-based models have been proposed for HSI classification, they face the following challenges.

1) *How to combine all spatial position information of the HSI cube in every feature extraction operation.* For an HSI cube with a complex background, the local features extracted by CNN or GCN often have different properties with the central pixels' category, and it prevents the model from recognizing the target pixel features leading to redundant repeated calculation. We believe that every feature extraction operation should make full use of the global spatial location features to avoid inappropriate feature extraction.
2) *How to make the network take the initiative to select useful spatial features and weaken the irrelevant spatial features for classifying the center pixel.* In an HSI cube, not all of the pixels have a strong correlation with the center pixel; many irrelevant pixels inevitably exist and these irrelevant pixels may be harmful to the classification of the HSI cube [32]. For example, in the Indian Pines (IP) dataset, class "oats" only includes 20 samples, and all of these samples exhibit narrow and long distributions; this classification is easily disturbed by surrounding pixels. In this condition, an HSI cube usually includes many irrelevant pixels, and the impact of irrelevant pixels will obscure the contribution of major pixels so that the classification results will likely be incorrect.
3) *How to explore all pixel labels in the HSI cube to guide the network to learn more accurate fitting parameters.* An HSI cube may contain multiple pixel categories, and the category of the center pixel may not accurately represent the characteristics of the HSI cube. Use of the central pixel as the training target is not sufficient in guiding the learning of deep model; many spatially detailed features and context correlation information are ignored, and the network should be more sensitive to the feature information in HSI cube center position. We believe that if the neural network not only focuses on the category labels but also pays attention to all of the pixels with the same category as the central pixel, the neural network can fully learn the object pixel characteristic and spatial detail information for a better fit of the object pixel features.

To address the abovementioned challenges, in this article, we propose an optimized transformer framework that combines spatial attention and a multibranch prediction

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approach (spectral spatial transformer (SST)-M). The transformer model [33] adopts a multihead self-attention (MHSA) mechanism to combine all node information in a sequence, and it is beneficial for improving the defect of the limited receptive field in CNN. The spatial sequence attention (SSA) method is proposed to adaptively learn the importance weight in each sequence location for classifying the center pixel, enhancing the contribution of useful spatial location features, and weakening the impact of irrelevant pixels in the HSI cube. To guide the network in learning exhaustive object pixel features, we relabel all the pixels of the HSI cube and add a mask prediction (MP) branch to classify all pixel properties. Correspondingly, the loss function of our network also contains two parts: the MP loss and classification prediction loss. The MP branch urges the network to pay attention to detailed spatial land-cover distributions in addition to the category label in the HSI cube to improve the problem of the insufficient feature fitting ability. However, there are still two issues that need to be addressed when migrating the transformer model to the HSI classification task. First, HSIs contain a considerable amount of redundant spectral information, and not all of the spectral bands will be beneficial for accurate classification [34]; this will give rise to high-computational complexity for the transformer model. Second, it is unclear how to mold the 3-D HSI cube into a sequence structure acceptable to the transformer model. To address the first question, a spectral feature extraction (SFE) net is proposed to select more discriminative spectral features and compress spectral dimensions. To address the second question, we use the method reported in [35], in which the input cube is organized into significant 3-D patches and reshaped into a patch sequence as the network input. The main contributions of our article are as follows:

1) We adopt the transformer encoder model as the backbone network that uses the MHSA structure to extract context features. All position information is considered in each self-attention (SA) layer. In addition, we used a two-layer convolution network to assemble more representative spectral features and compress the bands of HSI cubes.
2) We design a learnable SSA module that provides attention weight to represent the spatial location importance for classifying the HSI cube, implicitly enhancing useful spatial features and weakening irrelevant spatial features.
3) We design a MP branch to make the network pay attention to all pixel categories in the HSI cube; this branch is parallel to the classification predict branch and can be trained in an end-to-end manner. This module drives the network to fit the detailed spatial texture information and target pixel distribution characteristics in addition to the category label in the HSI cube.
4) We conduct many experiments on the IP, University of Pavia (PU), Kennedy Space Center (KSC), and Houston datasets. The obtained results demonstrate the effectiveness of the proposed SFE net, SSA module, and MP module and shows how the proposed method (SST-M) outperforms other leading methods.

The remainder of this article is organized as follows. Section II presents related work about HSI classification. Then, Section III describes the details of the proposed method. Section IV introduces the experimental setup and analysis of the results. Section V concludes this article.

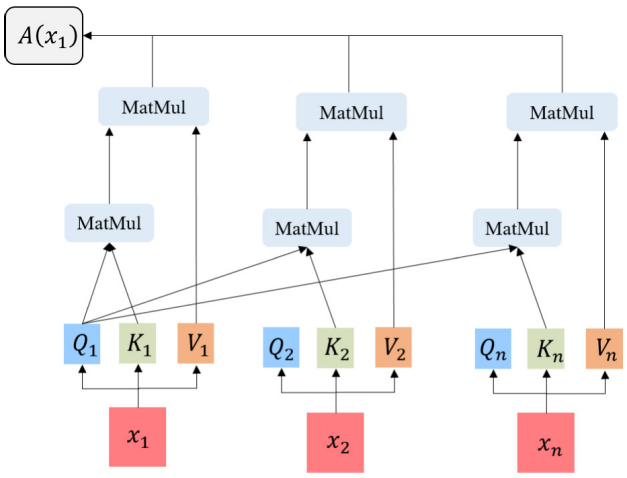


Fig. 1. SA model and the weight calculation flowchart.

II. Related Work

A. Attention Mechanism Model

Inspired by the human visual perception system, the attention mechanism has been widely used in DL in recent years. The channel attention mechanism was proposed in 2017. Hu *et al.* noticed that in a deep network, the importance of each channel of the feature map is different and proposed the Squeeze-and-Excitation networks (SE) model [36] to assign weights to each channel, allowing the information between the channels to interact. In visual tasks, the spatial attention mechanism is favored. The essence of the spatial attention mechanism is to locate the spatial information of interest and suppress useless information. The representative spatial attention mechanism methods include spatial transformer networks (STNs) [37]) and dynamic capacity networks (DCNs) [38]. Subsequently, convolutional block attention module (CBAM) [39] and dual attention network (DAN) [40] were developed; these methods simultaneously employ channel attention and spatial attention mechanisms to solve specific visual tasks. In the HSIC field, [22], [23] [34] are studies of fused attention mechanisms with CNNs for the extraction of the spatial and spectral features of HSIs.

The implementation of an attention mechanism relies heavily on external data, and the results are usually based on probability graphs or are displayed in the form of a probability feature vector. A SA mechanism was proposed to learn the internal connections between the data. Each node in a SA network can combine the information of all nodes to calculate an attention weight index to obtain richer contextual information and produce a long-distance memory ability. The model of the SA model is shown in Fig. 1. Each node will derive three feature vectors (query, key, and value). The calculation formula of the attention weight of a single node is as follows:

$$A(x_1) = \sum_{i=1}^{n} \left(\frac{Q_1 {K_i}^T}{\sqrt{d}} \right) V_i \tag{1}$$

$$\text{Attention}(x_1, x_2, \ldots, x_n) = \text{softmax}(A(x_1), A(x_2), \ldots, A(x_n)). \tag{2}$$

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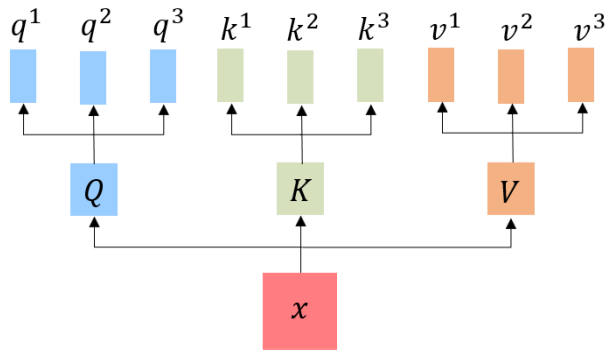


Fig. 2. MHSA model of one node.

Given $(x_1, x_2, \ldots, x_n)$ as the input sequence, n defines the sequence length of x. Q, K, and V are the abbreviations of query, key, and value, respectively, and d is the dimensions of Q and K. The scaled dot-product is used between all Q and K, and the softmax function is used to compute the attention weights for the sequence.

In the MHSA mechanism, the queries, keys, and values will be projected several times (p times). The feature expression form of a node is shown in Fig. 2. The calculation formula of the MHSA weight is as follows:

$$A(x) = \sum_{i=1}^{n} \sum_{j=1}^{p} \left(\frac{q^i (k^j)^T}{\sqrt{d}} \right) v_j \tag{3}$$

where d is the dimension of q and k. We used p heads such as $(q^1, q^2, \ldots, q^p)$.

The SA mechanism was also introduced into computer vision tasks; for example, Wang *et al.* proposed nonlocal neural networks [41] that include a nonlocal block that adapts the SA model and embeds it into convolution networks. The nonlocal block can capture the long-range dependence of time (1-D timing signal), space (picture), and space-time (video sequence). The self-attention generative adversarial network (SAGAN) [42] successfully introduces the SA mechanism into GANs and achieves better results. The traditional convolution method is a local information-sensitive operation, and long-distance dependencies can be obtained only through several convolution layers. This prevents the network from learning long-term interactions. By contrast, the SA model is designed from the perspective of comprehensive global information and has two obvious advantages: redistributing weights and obtaining global dependencies. In fact, the SA model can also be used as an independent network layer to replace CNN to extract high-level features.

B. Transformer Module in Computer Vision

In 2017, Vaswani *et al.* [33] first introduced the transformer module in the natural language processing (NLP) field and obtained state-of-the-art results. This model is mainly used to overcome the problem encountered in traditional neural networks of spending a large amount of time in the training phase and the difficulty of achieving parallel computing and therefore abandons the traditional CNN and RNN. The whole network structure is composed of a SA mechanism.

In computer vision (CV) tasks, Carion *et al.* [43] first apply the transformer model to the field of object detection and propose models named Bidirectional Encoder Representation from Transformers (BERT) that replace the manual design elements, e.g., anchor and region proposal network (RPN). However, the drawbacks of BERT are that it is difficult to train to convergence and exhibits poor performance in small object detection tasks. To solve these issues, Zhu *et al.* [44] associate deformable convolution and a transformer to explore the sparse spatial location attention method to improve the problem of slow convergence and high-computational complexity. The vision transformer (ViT) model was proposed in [31] and merges the transformer encoder in the image classification task. In this model, the images are divided into sequences of flattened 2-D patches. Then, patch embedding is used to unify the dimensions of the input sequence in the SA mechanism. Finally, the multilayer perceptron (MLP) module is used to obtain the classification result. Touvron *et al.* [45] optimizes the training strategy used in ViT and the proposed DeiT model. Compared with ViT, DeiT does not need a very large amount of training data and spends less time but still obtains better results. For HSI classification, HSI-BERT was proposed in [46], and flexible select diversity regions containing target pixels, including round, square, line, and joint training, were allowed in HSI-BERT to improve training efficiency. T-SST-L was proposed in [47], in which the author used CNN to extract the spatial features and a modified transformer to capture sequential spectral relationships. In [48], the densely sampled method was applied to divide the spectral channel into multiple groups, and then, the author used groupwise spectral embedding methods and a transformer encoder module to extract advanced features. In [49], MHSA modules were also used to extract the image features. Different from previous work, the author paid more attention to the spatial features in HSI. In [50], the author designs a spectral-spatial feature tokenization transformer method which delive the level features extracted by CNN to transformer network to capture high-level semantic features. This is a two-stage feature extraction network and which integrates the CNN and transformer structure.

Based on the comprehensive analysis above, we propose a novel model using the transformer encoder structure combined with spatial attention and a MP approach to solve HSIC.

III. Proposed Method

In this section, we introduce in detail the workflow of the SST-M method in this article. Fig. 3 shows the flowchart of the SST-M framework. For each target pixel, a sign cube that contains the target pixel is selected and used as the input of our models. The SFE net is first used to compress the spectral dimension of the HSI cube. The compressed cube is then split into a patch sequence, and each patch is reshaped into a 1-D vector. Meanwhile, the SSA module generates position weight

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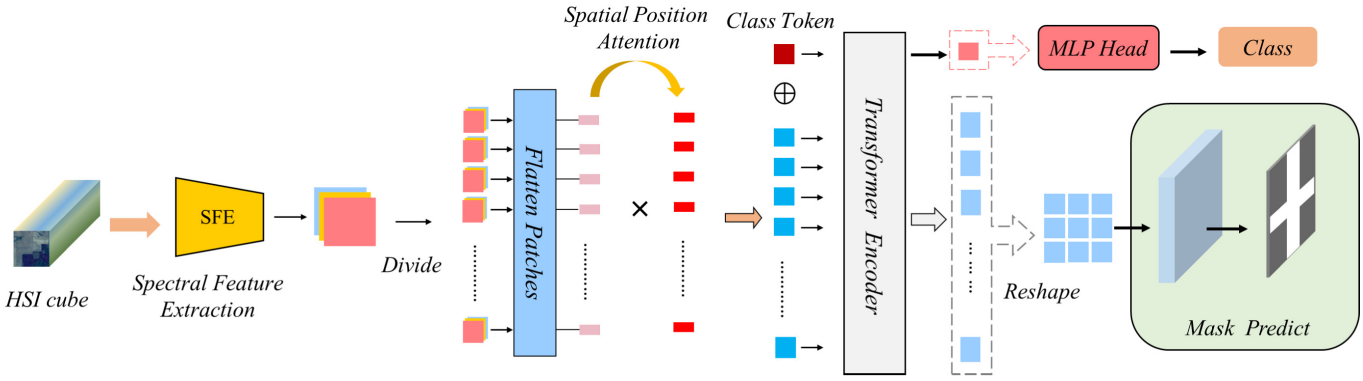


Fig. 3. Architecture of the proposed method (SST-M) for HSI classification. ⊕ denotes the vector concatenate operation, and × denotes that the feature vectors multiplied by the spatial weight. First, an HSI cube is compressed and partitioned into a patch sequence. A transformer encoder is used to extract abstract features. MLP is a two-layer connection network for the generation of classification results. The MP branch predicts all of the pixel category characteristics. The MP branch consists of a reshape operation, a transposed convolution layer, and a 1 × 1 convolution layer.

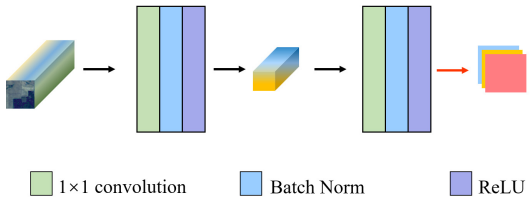


Fig. 4. Structure of the SFE net and the channels of the intermediate feature map are half of the input HSI cube, and the output feature images of this module only have three channels.

information from the input sequence. All of the patch vectors multiples the position weight, and the class token is fed into a SA network to extract advanced features. A class token is attached, and a MHSA mechanism is used to learn interrelated features. Finally, a fully connected neural network is used to obtain the classification result, and a deconvolutional network block is designed to predict the segmentation map according to the SA learned features.

A. SFE Network

HSIs have abundant spectral dimensions that include considerable amounts of redundant information. Each spectral band makes different contributions to different land-cover categories. Based on this assumption, we used a two-layer convolutional network to capture band features with positive contributions from the raw input cube called SFE net. In this network, we collect only the spectral information and preserve the spatial location information as much as possible. Fig. 4 shows the structure.

There are only two convolution blocks in the SFE net, and each block consists of a 1 × 1 size convolution layer, BatchNorm layer, and rectified linear unit (ReLU). Among these, the first convolution layer reduces the dimensions to half of the input HSI cube. In the second layer, only three convolution kernels are used; thus, the output spectral feature cube of the SFE net has only three channels.

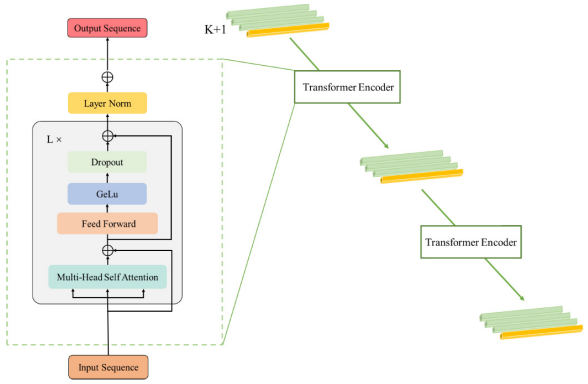


Fig. 5. Transformer Encoder module in this article.

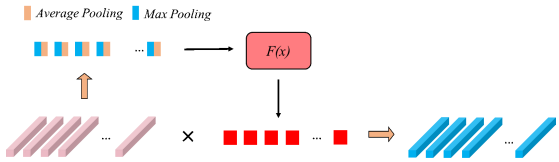


Fig. 6. Workflow of SSA model. It utilized max-pooling and average-pooling to sampling the feature vectors and transfer the pooling output as the sequence weight.

B. Transformer Classification and Position Attention

1) Divide and Flat: After the SFE net extracts spectral information, the size of the spectral features cube is $w \times w \times c$, and we divide the spectral feature cube $X \in \mathbb{R}^{w \times w \times c}$ into a patch sequence $x \in \mathbb{R}^{N \times (p^2 \times c)}$, where w is the width of the HSI cube, c is the number of channels and $c = 3$, p is the width of divided patches, and $N = (w/p)^2$. For each patch, the flatten operation is employed to reshape it into a 1-D vector.

It is appropriate to divide the patch into 3 or 5 so that each patch will contain neighborhood spatial information, and the patch sequence will not be too long, which is conducive to reducing the computational complexity of the transformer model.

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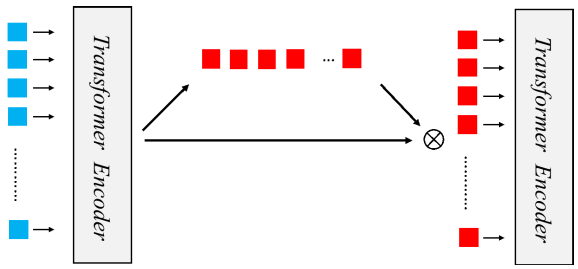


Fig. 7. SSA. Given input or intermediate features, spatial weight is computed and multiplied to them as the input of the following transformer block.

2) SSA Module: Numerous studies [35], [43] [44] have demonstrated the importance of position information for transformer models. In this article, the purpose of the spatial attention is to provide spatial weight information for the input sequence and make the network to select effective spatial features and weak invalid spatial features. The SSA method directly calculates the attention weight from the input sequence. Specially, the maximum pooling and average pooling are used to sample each feature vector in the sequence (4) and (5). The 1×1 convolution is used to calculate the attention weight from the sampling results (6) and the convolution parameters are self-learning by the network; then, the feature weights are multiplied by the corresponding feature vectors (7). The workflow of SSA is shown in the Fig. 6.

$$Sa_i{}^{\max} = \max(x_{i,j}) \tag{4}$$

$$Sa_i{}^{\text{ave}} = \frac{1}{c}\sum_{j=1}^{c} x_{i,j} \tag{5}$$

$$Sa_i^* = F\left(\left[Sa_i{}^{\max}, Sa_i{}^{\text{ave}}\right]\right) \tag{6}$$

$$x_i{}^{\text{new}} = x_i * Sa_i^* \tag{7}$$

where $i \in [1, \text{len(sequence)}]$, $j \in [1, c]$ and the len(sequence) represent the length of the input sequence, and c represents the dimension of each feature vector in input sequence. The SSA method is used between each transformer block, as shown in Fig. 7.

3) Transformer Classification: The architecture of a single-layer transformer encoder module in this article is shown in Fig. 5. The principle of the transformer module is a multiheaded SA block, the position weight is multiplied to patch sequences, and the class token is concatenated to patch sequences, as shown in (8). We note that SFE net has boosted feature diversity from the HSI cube so that the trainable linear projection operation described in [35] can be abandoned in our network. LayerNorm (LN) is used after the encoder module, and residual connections are applied between adjacent SA blocks and fed forward connections (9) and (10); spatial weight is computed and multiplied to the output as the input of the following transformer block (11). A MLP is used to achieve the classification task according to the class token (12). In this article, MLP contains two layers of a fully connected neural network, and the number of output neural networks is equal to the number of classes

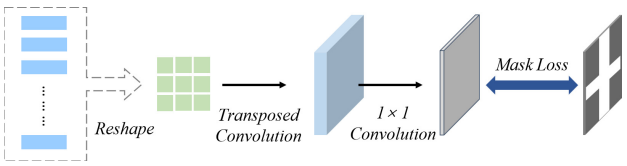


Fig. 8. MP branch consists of a reshape operation, a transposed convolution, and a 1×1 convolution. The mask loss is also added to guide learning.

$$z_0 = x_{\text{class}} \oplus \left(\left[x_p^1; x_p^2; \ldots; x_p^N\right] * E_{\text{position}_0}\right) \tag{8}$$

$$z_l = \text{SA}(z_{l-1}) + z_{l-1}, \quad l \in [1, L] \tag{9}$$

$$z_f = F(z_l) + z_l, \quad l \in [1, L] \tag{10}$$

$$z_f = z_f * E_{\text{position}_l}, \quad l \in [1, L] \tag{11}$$

$$y = \text{MLP}\left(\text{LN}\left(z_f^0\right)\right) \tag{12}$$

where z_0 is the input sequence and E_{position} is the spatial sequence weight. x_{class} is the class token, x_p^N is the pth patch, and $F(\cdot)$ is the forward functions. SA$(\cdot)$ is the SA layer, z_l is the output of the lth SA layer, z_f is the output of the lth feed forward, and L presents the number of SA blocks. MLP$(\cdot)$ is the classification prediction network, z_f^0 is the first node in the output sequence, and LN$(\cdot)$ is the layer normalization function.

C. MP Approach

1) Mask Label Generate: A new HSI cube labeling method is proposed to obtain mask labels; all of the pixels in the HSI cube are relabelled to adequately represent the characteristics of the HSI cube. Specifically, if the pixels in the HSI cube have different land-cover categories from the center pixels, we label "0" at the corresponding spatial location representing background areas; pixels have land-cover categories that are identical to center pixels and the original label information remains. The mask label shows the spatial distributions and pixel characteristics of the object category.

2) MP Branch: In the transformer encoder, only the class token is used in MLP to generate classification results, and the information of the class token depends on other nodes in the feature learning process. The purpose of the mask branch is to enhance the feature expression ability for all nodes to improve classification accuracy. Inspired by instance segmentation, we construct a MP branch parallel to MLP to predict all of the pixels belonging to the object category or background. As shown in Fig. 8, the remainder of the node sequence other than the class token is reshaped into a 2-D patch, and then a transposed convolution layer and a 1×1 size convolution layer are used to obtain the final prediction map. The function of the MP branch can be described by (13)

$$\text{mask} = \text{MASK}\left(\text{BN}\left(z_{\text{out}}^{1-N}\right)\right) \tag{13}$$

where z_{out}^{1-N} represent the other nodes except z_{out}^0 in the output sequence. MASK$(\cdot)$ is the MP function, and BN$(\cdot)$ is the batch normalization function.

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3) Training and Training Loss: Similarly, the overall loss function consists of two parts: the classification prediction loss and the MP loss, as follows:

$$L(p, c, m, m^*) = \alpha L_{\text{cla}}(p, c) + (1 - \alpha) L_{\text{mask}}(m, m^*) \quad (14)$$

where L_{cla} is the classification cross-entropy loss, L_{mask} is the binary mask cross-entropy loss, p is the classification prediction results, c is the classification ground truth, m is the output from the mask feedback branch, represents the MP results, and m^* is the ground-truth map. α is the balance factor in L_{cla} and L_{mask}.

IV. Experiments and Results

To estimate the effectiveness of SST-M, three groups of experiments are set up on three mainstream HSI datasets. The first experiment verifies the influence of the network depth and the number of SA heads. The second set of experiments consists of ablation experiments that aim to validate the functionality of the SFE net, SSA module, and MP branch. The last are contrast experiments that compare the class performance of SST-M with other leading HSI classification methods.

A. Datasets

In this section, four leading HSI datasets are selected to conduct HSI classification experiments, including IP, PU, KSC, and Houston. Approximately, 10% of the samples are randomly selected as the training set for IP, PU, and KSC, and 5% of the samples are taken for Houston, and the rest of the samples are taken as the test set. Table I–IV shows the specific details of the data division.

1) Indian Pines (IP): IP were first used for HSI classification and were gathered by an Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor in an Indian pine forest in northwestern Indiana in 1992. The IP data have 145 × 145 pixels and 220 spectral bands, but because the 104–108, 150-163, and 220 bands are not reflected by water, we generally use the remaining 200 bands as the research object after eliminating these 20 bands. There are a total of 16 classes, but the distribution of them throughout the samples is very uneven.

2) Pavia University (PU): The PU database was collected in the Italian city of Pavia in 2003. The PU data contain 610 × 340 pixels, and the spatial resolution of each pixel is 1.3 m. The spectral image has 115 continuous bands in the 0.43–0.86 μm wavelength range, but 12 bands are eliminated due to the influence of noise, so that the image formed by the remaining 103 spectral bands is generally used. In total, nine classes are contained in the PU database.

3) Kennedy Space Center (KSC): The KSC database was collected by a National Aeronautics and Space Administration (NASA) AVIRIS instrument over the KSC, Florida, in 1996. The KSC data contain 512 × 614 pixels, and the spatial resolution of each pixel is 18 m. The number of available spectral bands is 176 after removing water absorption and low signal-to-noise ratio (SNR) bands. There are 13 classes contained in the KSC database.

TABLE I
Land-Cover Category and Data Division on IP Dataset

Class No.	Class Name	Training	Test
1	Alfalfa	5	41
2	Corn-notill	143	1285
3	Corn-mintill	83	747
4	Corn	24	213
5	Grass-pasture	49	434
6	Grass-trees	73	657
7	Grass-pasture-mowed	3	25
8	Hay-windrowed	48	430
9	Oats	2	18
10	Soybean-notil	98	874
11	Soybean-mintill	246	2209
12	Soybean-clean	60	533
13	Wheat	21	184
14	Woods	127	1138
15	Buildings-Grass-Trees-Drives	39	347
16	Stone-Steel-Towers	10	83
Total		1031	9218

TABLE II
Land-Cover Category and Data Division on the Pavia University Dataset

Class No.	Class Name	Training	Test
1	Asphalt	664	5967
2	Meadows	1865	16784
3	Gravel	210	1889
4	Trees	307	2757
5	Painted metal sheets	135	1210
6	Bare Soil	503	4526
7	Bitumen	133	1197
8	Self-Blocking Bricks	369	3313
9	Shadows	95	852
Total		4281	38495

4) Houston: The Houston 2013 database was collected by the National Center for Airborne Laser Mapping (NCALM) at the University of Houston (UH) in June 2012. The data were provided by the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) data fusion competition. The data contain 349 × 1905 pixels, including 144 spectral bands ranging from 364 to 1046 nm. There are 13 classes contained in the Houston 2013 database.

B. Analysis of Model Depth and Attention Heads

Two sets of experiments are performed to validate the effect of model depth and attention heads on classification results. The MP branch is removed to accelerate network convergence in these experiments. Specifically, the model depth is the number of SA layers in our model. First, we define the model depth as 3 and the number of heads as 2, 4, 6, and 8. Second, we fix the number of attention heads to 6 and vary the model depth from 2 to 4.

In all of these experiments, the square region is 15 × 15 and the learning rate is set to 0.01. We use the overall accuracy (OA) as the evaluation indicator in these experiments, and the results obtained after 500 training epochs for IP and KSC dataset, 200 epochs for PU dataset are listed in Tables V and VI.

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TABLE III
LAND-COVER CATEGORY AND DATA DIVISION ON THE KSC DATASET

Class No.	Class Name	Training	Test
1	Scrub	77	684
2	Willow swamp	25	218
3	Cabbage palm hammock	26	230
4	Cabbage palm/oak hammock	26	226
5	Slash pine	17	144
6	Oak/broadleaf hammock	23	206
7	Hardwood swamp	11	94
8	Graminoid marsh	44	387
9	Spartina marsh	52	468
10	Cattail marsh	41	363
11	Salt marsh	42	377
12	Mud flats	51	452
13	Water	93	834
Total		528	4683

TABLE IV
LAND-COVER CATEGORY AND DATA DIVISION ON THE HOUSTON DATASET

Class No.	Class Name	Training	Test
1	Healthy grass	76	1436
2	Stressed grass	80	1520
3	Synthetic grass	44	831
4	Trees	70	1321
5	Soil	71	1349
6	Water	19	354
7	Residential	81	1543
8	Commercial	75	1415
9	Road	86	1624
10	Highway	78	1489
11	Railway	86	1637
12	Parking Lot 1	79	1493
13	Parking Lot 2	35	661
14	Tennis Court	28	537
15	Running Track	44	834
Total		950	18054

An examination of the results shows that the best number of model layers is 3 for all three datasets. For Indian Pines, the best number of attention heads is 6; for Pavia University, the best number of attention heads is 8; and for KSC, the best number of attention heads is 4. Classification ability remained relatively stable when the model depth and number of heads changed in the KSC and PU datasets; however, for the IP dataset, the classification performance changed greatly with different model depths. It is clear that it is difficult to fit the category features in the IP dataset because of the extremely unbalanced data distribution.

C. Ablation Experiments and Results Analysis

1) Experimental Setup: In this section, we conduct three ablation experiments to prove the significance of the SFE net, SSA module, and MP branch.

Specifically, in the first experiment, we use the principal component analysis (PCA) algorithm to replace the SFE net to compress the spectral information. For fair comparison, the spectral channels were compressed to three dimensions, and the MP branch was removed. The aim of this experiment is to compare the classification results of SFE and PCA.

In the second experiment, we replace the SSA module with a predefined position embedding matrix in [31] and test its classification performance to show the effectiveness of the SSA. Similarly, the MP branch is not used in this experiment.

The third experiment compares the classification results of our method with and without the MP branch. The benchmark network only uses the SFE net and SSA model, and the final network includes the SFE, SSA, and MP branches. In the final network, the loss function is shown in formula (9), where $\alpha = 0.8$. Table VII presents the classification results with different configurations in three datasets.

2) Results Analysis: Lines 2 and 4 in Table VII show the results of PCA and SFE, respectively. Our proposed SFE net achieves higher class performance in the OA index than PCA. Our proposed SFE net demonstrates better feature extraction ability because it is a supervised learning module and can be trained alongside the network. Only a two-layer convolutional neural network is used in SFE and achieves an absolute improvement of more than 1.0% compared to PCA on IP and PU; therefore, the computational resource consumption is acceptable.

Lines 3 and 4 in Table VII show the results for examining how the approach used to obtain position information affects the classification performance. Our proposed SSA method obtains 98.08%, 98.51%, and 98.57% accuracy scores, and the conventional method without SSA obtains 95.78%, 98.16%, and 96.12% scores. Our module generates more appropriate position attention features based on spatial sequence weight. This demonstrates that not all sequence features provide the same contribution on transformer block, and the SSA model

TABLE V
OA RESULTS OF EXPERIMENTS WITH DIFFERENT NUMBERS OF HEADS

Number of Heads	IP	PU	KSC
2	97.20	97.36	97.64
4	97.54	98.29	99.03
6	98.68	98.51	98.57
8	98.33	99.10	97.89

TABLE VI
OA RESULTS OF EXPERIMENTS WITH DIFFERENT MODEL DEPTHS

Number of Layers	IP	PU	KSC
2	97.62	98.17	98.55
3	98.68	98.51	98.57
4	97.20	97.95	98.32

TABLE VII
CLASSIFICATION RESULTS OF THE PROPOSED METHOD WITH DIFFERENT CONFIGURATIONS

Dataset	IP	PU	KSC
PCA + SSA	96.55	97.41	-
SFE	95.78	98.16	96.12
SFE + SSA	98.08	98.51	98.57
SFE + SSA + MF	99.08	99.61	99.70

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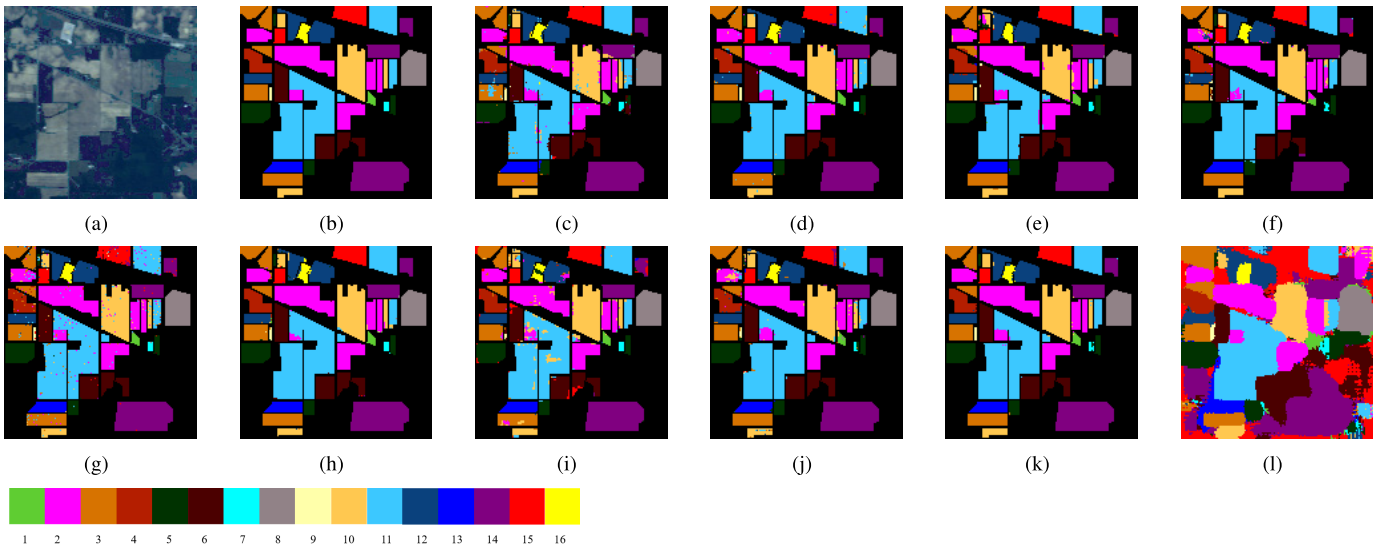


Fig. 9. Classification maps for the IP with compared methods. (a) Source Image. (b) Ground Truth. (c) 2-D-CNN. (d) SSAN. (e) DAGCN. (f) HSI-BERT (g) T-SST-L (h) SSFTT. (i) SSRN. (j) RSSAN. (k) SST-M. (l) All pixels classification map by SST-M.

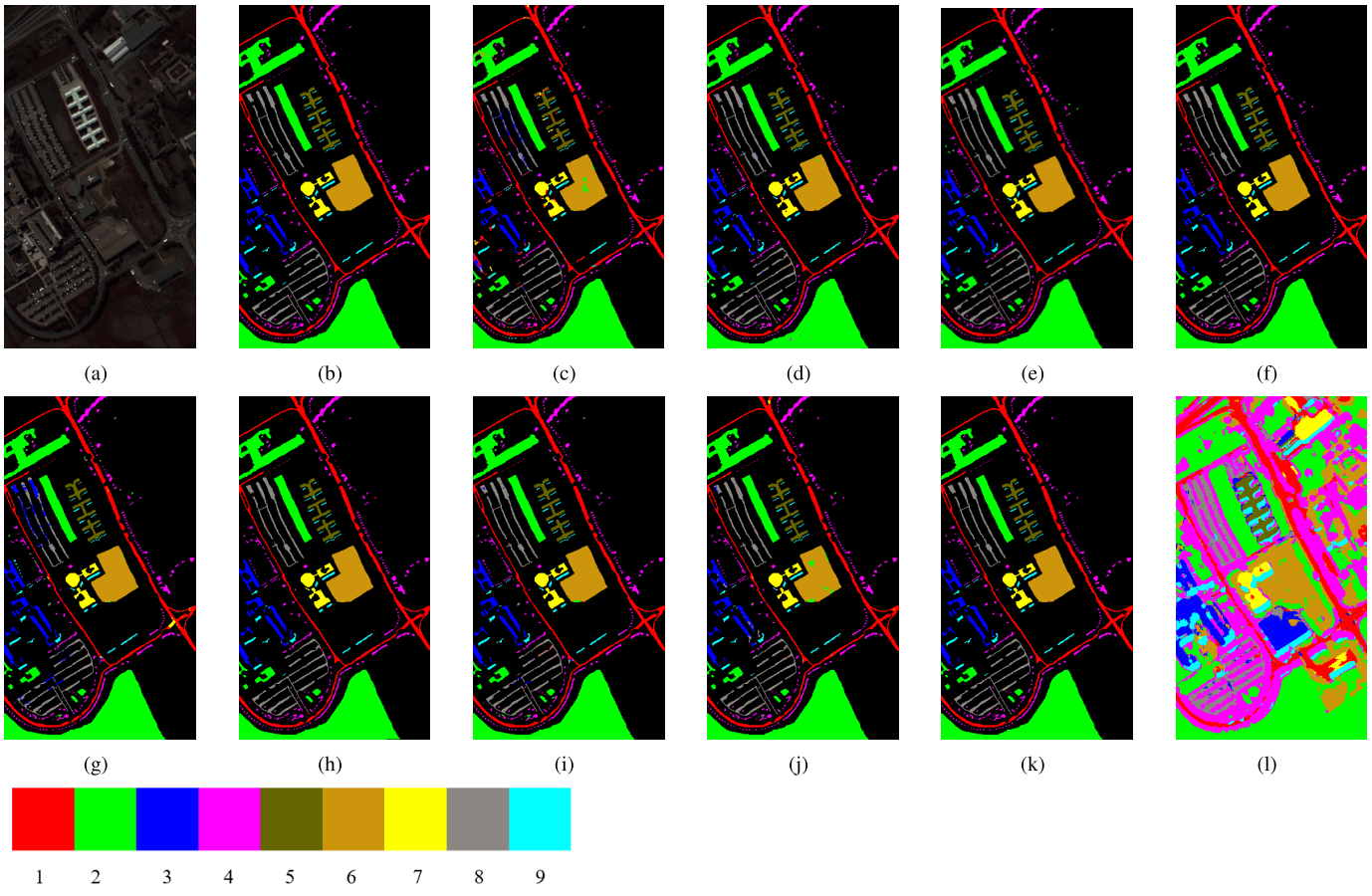


Fig. 10. Classification maps for the PU with compared methods. (a) Source Image. (b) Ground Truth. (c) 2-D-CNN. (d) SSAN. (e) DAGCN. (f) HSI-BERT. (g) T-SST-L. (h) SSFTT. (i) SSRN. (j) RSSAN (k) SST-M. (l) All pixels classification map by SST-M.

adaptively find feature vectors that are useful for HSIC and implement it in an end-to-end network.

Lines 4 and 5 in Table VII show the results of adding the MP branch. For the IP, PU, and KSC dataset, the classification result with the MP branch is apparently improved; By comparison, in IP and KSC, only a small number of samples are available to train the network, and the distribution of the samples is very uneven. This demonstrated that our MP method enhances the fitting ability of the neural network and improves the quality of the parameters.

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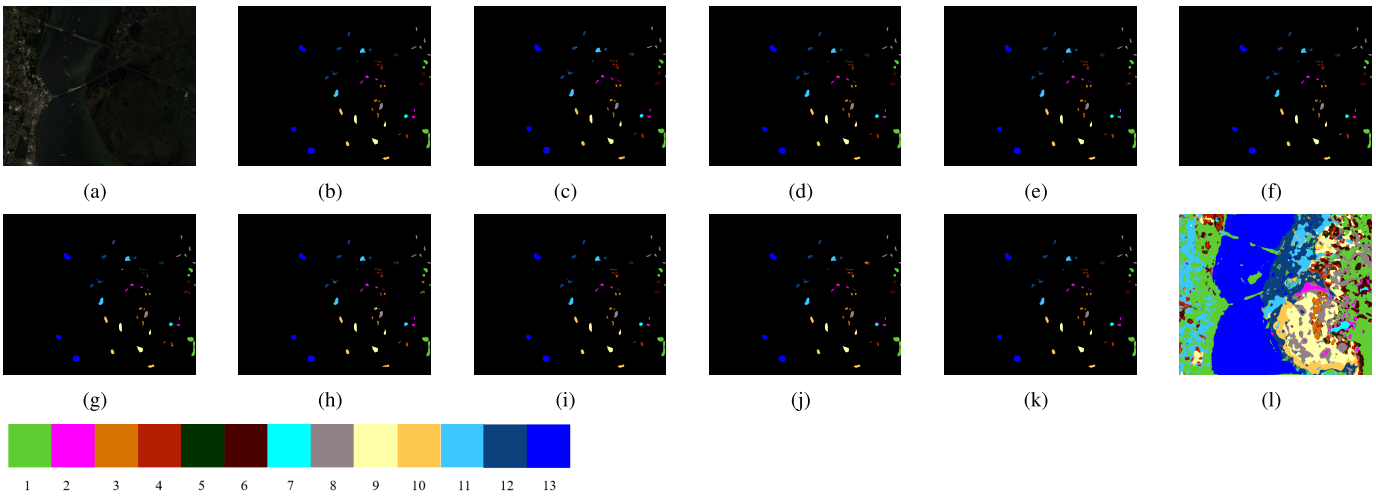


Fig. 11. Classification maps for the KSC with compared methods. (a) Source Image. (b) Ground Truth. (c) 2-D-CNN. (d) SSAN. (e) DAGCN. (f) HSI-BERT. (g) T-SST-L. (h) SSFTT. (i) SSRN. (j) RSSAN. (k) SST-M. (l) All pixels classification map by SST-M.

TABLE VIII
CLASSIFICATION RESULTS OF OUR PROPOSED METHOD AND OTHER LEADING METHODS ON THE IP DATASET (%)

Class No.	2D-CNN	SSAN	DAGCN	HSI-BERT	T-SST-L	SSFTT	SSRN	RSSAN	SST-M
1	75.61±19.51	92.85±2.19	92.78±4.21	72.68±0.17	79.31±1.39	99.27± 2.19	**100±0.00**	87.10±4.32	99.27±1.12
2	88.85±3.78	97.39±0.77	94.34±1.59	96.06±0.02	84.76±7.47	95.24± 0.42	98.12±0.01	90.89±1.37	**98.59±0.81**
3	92.58±0.67	97.35±1.57	96.68±2.34	97.62±0.01	94.12±6.26	98.53± 0.65	98.46±0.01	90.88±1.06	**99.21±0.50**
4	95.15±2.58	96.48±1.65	97.56±2.03	97.28±0.02	95.65±4.31	**99.67± 0.67**	97.04±0.02	81.82±7.50	96.95±0.90
5	88.47±6.98	98.55±0.75	95.95±2.43	97.51±0.01	87.10±4.21	98.74± 0.66	97.16±0.03	98.81±0.46	**98.89±0.59**
6	94.48±2.21	**99.91±0.13**	98.42±0.39	81.37±5.69	98.69±0.01	99.79± 0.23	98.88±0.05	98.43±0.56	99.33±0.51
7	84.94±10.09	97.65±3.22	90.00±8.24	62.40±0.20	**100±0.00**	99.60± 1.20	41.94±0.33	94.74±2.39	**100±0.00**
8	97.09±2.96	**100±0.00**	99.81±0.42	**100±0.00**	97.42±10.2	99.95±0.09	99.89±0.08	98.50±2.23	**100±0.00**
9	79.65±23.51	86.67±9.50	94.00±13.42	38.89±0.18	78.95±1.13	68.33±17.40	33.33±0.42	71.43±4.97	**100±0.00**
10	85.01±12.67	96.36±0.63	92.59±2.62	97.21±0.02	96.53±0.89	97.27± 0.69	**99.16±0.01**	94.40±1.78	98.55±1.52
11	94.28±2.44	98.53±0.58	98.21±0.45	99.35±0.00	98.11±0.77	99.33± 0.31	86.37±0.01	97.73±1.34	**99.52±0.62**
12	91.38±4.68	96.35±1.36	96.36±2.19	95.08±0.02	89.75±1.67	96.95±1.50	79.73±0.02	93.72±2.27	**96.96±1.10**
13	96.91±3.12	99.51±0.44	99.46±1.22	98.80±0.02	90.50±1.45	99.24±1.04	98.53±0.02	100.00±0.00	**100±0.00**
14	97.25±2.73	99.37±0.54	99.12±1.01	99.51±0.00	91.44±2.00	**99.92± 0.07**	99.74±0.03	99.21±0.18	99.85±0.10
15	96.81±2.62	98.54±0.95	97.41±1.86	99.14±0.01	99.15±1.28	98.29±0.86	89.55±0.05	87.73±5.47	**99.63±0.59**
16	80.17±8.80	**98.21±1.26**	90.12±8.37	91.81±0.05	81.54±2.45	80.96±6.02	86.24±0.06	95.31±2.80	97.35±1.69
OA	92.48±2.10	98.33±0.47	96.86±0.36	97.75±0.00	91.55±1.82	98.21±0.17	94.16±0.01	95.17±0.76	**99.08±0.31**
AA	89.91±2.89	97.04±1.12	95.80±0.87	90.13±0.02	90.41±2.54	95.69±1.21	92.67±0.01	92.54±1.87	**99.01±0.33**
Kappa	91.62±2.47	97.87±0.5	96.42±0.41	97.43±0.01	89.91±1.66	97.96±0.19	93.37±0.06	94.49±1.99	**98.95±0.36**

D. Comparison With Other Algorithms

The last experiment compares the performance of SST-M with leading methods, including 2-D-CNN [11], spectral spatial attention network (SSAN) [23], DAGCN [25], HSI-BERT [46], T-SST-L [47], spectral spatial feature tokenization transformer (SSFTT) [50], spectral spatial residual network (SSRN) [51], and residual spectral spatial attention network (RSSAN) [34].

The OA, average accuracy (AA), and kappa coefficients (kappa) are used to assess the effectiveness of the proposed method on four standard datasets. The spatial size of the HSI cube is 15 × 15, the batch size is 128, the patch size is 3 × 3, the length of the input sequence for transformer encoding is 1 × 25, and the depth of the network and the number of attention heads are 3 and 6, respectively. The average result and standard deviation are recorded after ten runs.

The specific classification results for each algorithm using the IP, PU, KSC, and Houston datasets are shown in Tables VIII-XI. The colorful pixel classification result maps are shown in Figs. 9–12. Our method achieves better classification results for almost all land-cover categories. In particular, for the classes “Alfalfa”, “Grass-pasture-mowed”, “oats,” and “Stone-Steel-Towers” in the IP dataset, where only few samples are used for training and all pixels are located at the edge area of the categories, our method obtains classification results of 99.27%, 100%, 100%, and 97.35%, respectively; HSI-BERT obtains 72.68%, 62.40%, 38.89% and 91.81%, respectively; SSAN achieves 92.85%, 97.65%, 86.67% and 98.21%, respectively; 2-deep convolutional neural network (DCNN) achieves 75.61%, 84.94%, 79.65% and 80.17%, respectively; T-SST-L achieves 79.31%, 100%, 78.95% and 81.54%, respectively; and SSFTT achieves 99.27%, 99.60%, 68.33% and

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TABLE IX

CLASSIFICATION RESULTS OF OUR PROPOSED METHOD AND OTHER LEADING METHODS ON THE PAVIA UNIVERSITY DATASET (%)

Class No.	2D-CNN	SSAN	DAGCN	HSI-BERT	T-SST-L	SSFTT	SSRN	RSSAN	SST-M
1	98.56±0.58	99.06±0.18	99.18±0.22	**99.90±0.01**	96.43±1.33	99.64±0.25	99.85±0.02	99.16±0.75	99.82±0.20
2	99.01±0.45	99.73±0.11	99.75±0.12	**100±0.00**	98.45±1.41	99.91 ±0.14	99.97±0.02	99.36±0.23	99.99±0.01
3	87.26±5.36	96.21±0.69	97.66±1.06	**99.63±0.01**	92.59±4.85	99.13±0.51	97.26±0.03	95.17±3.54	98.77±1.72
4	92.58±1.25	98.47±0.91	95.92±0.83	**99.08±0.02**	96.48±2.25	97.50±0.79	97.19±0.02	98.09±0.24	98.79±0.5
5	99.21±0.66	**100±0.00**	98.32±1.04	**100±0.00**	99.17±0.16	98.74±0.74	99.53±0.01	99.36±0.25	**100±0.00**
6	95.36±1.19	99.14±0.55	100±0.00	99.99±0.00	91.45±1.38	99.99±0.02	**100±0.00**	99.43±0.75	**100±0.00**
7	91.17±5.78	95.88±1.66	98.90±0.86	99.98±0.00	96.57±1.03	99.67±0.51	99.71±0.01	94.95±1.28	**100±0.00**
8	98.00±0.99	97.31±0.50	97.52±0.99	**99.76±0.01**	97.04±2.88	98.78±0.55	99.19±0.05	96.55±1.19	98.83±1.54
9	85.43±6.39	99.53±0.64	92.36±1.97	**99.81±0.01**	97.89±8.34	95.08±1.76	99.46±0.06	99.40±0.38	98.22±3.35
OA	97.18±0.44	98.97±0.09	98.89±0.08	99.17±0.00	96.72±0.96	99.26±0.06	99.52±0.01	98.65±0.31	**99.61±0.28**
AA	94.89±0.57	98.37±0.14	97.73±0.15	**99.79±0.00**	96.23±1.25	98.13±0.16	99.13±0.01	97.93±0.56	99.23±0.65
Kappa	95.66±0.50	98.63±0.11	98.52±0.11	99.05±0.00	95.64±1.38	99.03±0.08	99.36±0.02	98.22±0.45	**99.48±0.37**

TABLE X

CLASSIFICATION RESULTS OF OUR PROPOSED METHOD AND OTHER LEADING METHODS ON THE KSC DATASET (%)

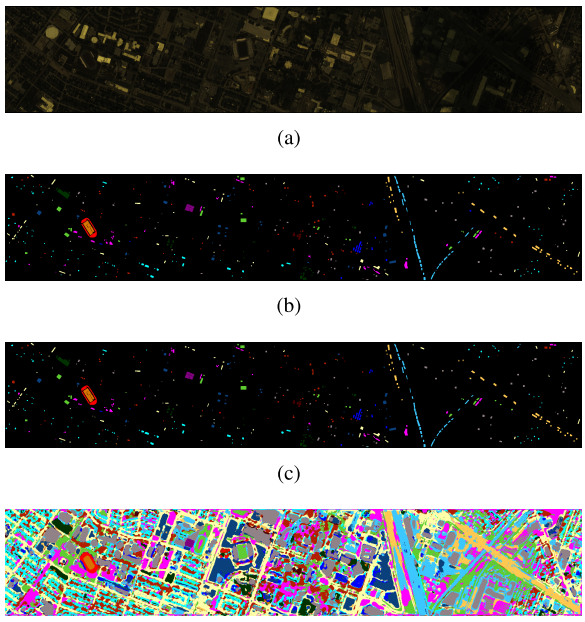
Class No.	2D-CNN	SSAN	DAGCN	HSI-BERT	T-SST-L	SSFTT	SSRN	RSSAN	SST-M
1	94.66±2.18	99.25±0.61	97.04±1.16	99.94±0.01	99.56±2.83	99.50±0.33	**100±0.00**	98.61±0.50	**100±0.00**
2	90.41±4.87	97.51±1.58	92.81±7.18	99.63±0.01	87.61±6.67	88.51±1.12	94.00±0.17	88.05±9.29	**100±0.00**
3	90.95±2.89	99.40±0.69	97.64±1.36	89.48±0.08	96.96±74.78	62.00±5.75	90.99±0.01	90.07±4.33	**100±0.00**
4	73.92±5.46	91.86±3.74	93.96±4.56	76.99±0.05	74.78±1.91	68.81±5.83	**95.26±0.03**	91.23±5.89	95.18±1.51
5	65.10±6.85	88.84±5.22	82.52±17.36	83.19±0.07	56.94±1.16	85.12±4.76	97.66±0.28	89.79±3.19	**98.89±1.13**
6	64.20±2.27	95.69±2.30	90.05±5.10	98.06±0.01	84.95±0.03	71.37±3.71	98.84±0.93	85.75±5.76	**99.95±0.15**
7	87.70±5.45	99.73±0.61	91.61±9.87	**100±0.00**	**100±0.05**	65.00±6.44	89.86±0.87	93.57±6.86	98.62±0.96
8	93.32±2.70	99.29±0.19	97.77±1.70	99.33±0.00	87.08±2.33	93.62±2.60	**100±0.00**	93.03±3.27	**100±0.00**
9	95.80±2.63	**100±0.00**	99.32±1.06	**100±0.00**	99.15±0.60	84.27±7.05	**100±0.00**	98.35±0.83	**100±0.00**
10	92.60±2.36	99.46±0.17	99.67±0.74	**100±0.00**	98.90±2.30	94.95±1.34	**100±0.00**	87.20±6.45	**100±0.00**
11	97.30±1.72	99.93±0.15	99.73±0.27	**100±0.00**	99.20±1.65	98.99±1.60	96.32±0.02	98.23±1.32	**100±0.00**
12	91.50±4.26	99.54±0.624	98.14±1.90	99.96±0.00	93.14±2.83	93.30±1.96	**100±0.00**	96.74±3.80	99.98±0.07
13	99.11±0.80	**100±0.00**	99.98±0.05	**100±0.00**	**100±0.00**	99.99±0.04	**100±0.00**	97.69±1.23	**100±0.00**
OA	91.40±0.54	98.71±0.51	97.20±0.57	97.69±0.00	94.04±0.97	93.82±4.76	98.41±0.02	93.04±0.51	**99.70±0.06**
AA	87.40±1.11	97.84±0.90	95.40±0.17	95.89±0.04	90.64±0.83	89.60±8.46	97.15±0.02	90.33±0.62	**99.43±0.01**
Kappa	90.46±0.62	98.48±0.49	96.89±0.64	97.42±0.01	93.36±1.09	93.10±5.38	98.23±0.02	93.30±1.71	**99.67±0.07**

TABLE XI

CLASSIFICATION RESULTS OF OUR PROPOSED METHOD AND OTHER LEADING METHODS ON THE HOUSTON DATASET (%)

Class No.	2D-CNN	SSAN	DAGCN	HSI-BERT	T-SST-L	SSFTT	SSRN	RSSAN	SST-M
1	97.25±1.77	93.83±4.53	98.67±0.09	96.73±0.02	97.65±10.13	95.05±1.49	98.86±1.61	96.02±0.97	**99.83±0.79**
2	**99.56±0.60**	94.34±5.33	99.36±0.04	98.37±0.01	98.62±4.95	98.83±1.18	98.20±3.24	98.68±0.66	99.47±0.09
3	99.87±0.19	98.21±1.15	99.72±0.00	99.30±0.01	98.18±10.59	99.02±0.84	99.71±0.5	**100±0.00**	**100±0.00**
4	94.88±7.63	96.97±2.57	94.37±0.32	97.13±0.02	94.81±4.05	98.11±1.06	**99.09±1.53**	96.65±0.79	98.95±1.10
5	99.94±0.14	99.82±0.25	99.69±0.14	99.63±0.01	98.20±2.17	99.90±0.19	99.55±0.50	98.90±0.76	**100±0.00**
6	88.66±6.89	92.12±6.96	99.08±0.15	74.60±0.08	87.87±0.18	99.94±0.12	**99.80±0.33**	85.49±2.77	93.28±0.94
7	92.62±2.58	96.19±2.61	98.81±0.08	92.19±0.03	90.29±3.36	90.22±0.80	96.50±1.51	91.86±2.50	**99.67±0.20**
8	88.55±3.73	92.66±3.88	73.95±0.47	88.60±0.02	88.51±2.23	95.53±2.36	96.95±2.90	87.07±1.71	**97.33±0.97**
9	93.35±3.68	96.36±2.77	86.95±1.49	94.38±0.03	94.35±3.67	97.55±0.99	97.33±1.33	88.94±3.18	**98.38±0.79**
10	99.00±1.10	97.65±6.62	97.01±1.78	94.10±0.03	95.32±0.06	99.65±0.44	98.44±1.36	94.84±1.48	**99.90±0.13**
11	94.47±2.36	95.85±3.14	95.31±1.40	93.79±0.02	95.81±4.15	98.53±0.49	99.09±1.68	94.43±1.52	**99.67±0.29**
12	96.51±2.01	96.85±2.92	97.27±0.59	93.43±0.02	94.01±3.17	**99.06±0.64**	97.46±1.17	90.38±4.58	98.87±0.42
13	95.28±1.80	91.02±3.52	90.97±1.13	82.77±0.06	86.09±4.47	**99.16±1.61**	90.71±3.46	82.13±5.47	95.72±0.91
14	99.57±0.91	**100±0.00**	99.26±0.36	93.86±0.04	99.57±4.84	99.86±0.18	99.94±0.13	97.71±1.38	**100±0.00**
15	99.84±0.23	99.63±0.61	**100±0.00**	99.60±0.01	99.44±3.80	99.92±1.12	99.46±0.42	98.89±0.75	99.64±0.33
OA	95.94±0.87	96.16±1.49	94.81±0.11	94.34±0.01	94.81±0.90	97.52±0.24	97.98±0.93	93.76±0.91	**98.80±0.13**
AA	95.81±0.00	96.10±1.49	95.36±0.12	93.23±0.01	94.58±1.38	97.82±0.19	98.00±0.80	93.47±1.21	**98.51±0.16**
Kappa	96.32±0.01	95.84±1.61	94.38±0.12	93.88±0.07	94.39±1.10	97.32±0.26	97.81±1.01	93.25±1.77	**98.71±0.15**

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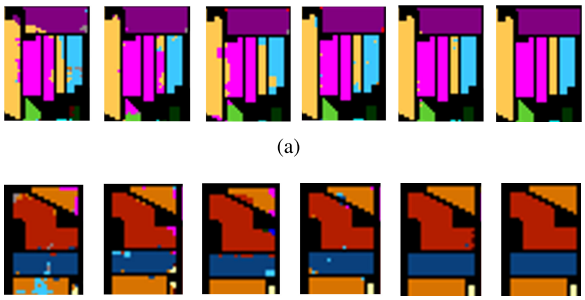


(a) (b) (c) (d)

Fig. 12. Experimental classification result on Houston dataset. (a) Source Image. (b) Ground Truth. (c) SST-M. (d) All pixels classification map by SST-M.

80.96%, respectively; SSRN achieves 100%, 41.93%, 33.33% and 86.24%, respectively; RSSAN achieves 87.10%, 94.74%, 71.43% and 95.31%. Some visualization classification results of local areas on IP datasets also show that our method obtains better classification results in edge areas, as shown in Fig. 13. It can be clearly seen that our proposed method has the least number of misclassified pixels. Compared with CNN-based and GCN-based methods, our proposed method possesses a wider receptive field and mask guide training; thus, only three MHSA layers are used in the network to achieve more accurate classification. Compared with HSI-BERT, our proposed fused attention mechanism precisely extracts the useful spatial location features and avoids the interference of incorrect pixels. The spectral features that were fully extracted also played an important role in improving the classification performance in our proposed method. Compared with SSRN and SSFTT, the mask feedback improves the ability of the model to extract target features. In addition, the T-SST-L, SSFTT and RSSAN methods has poor performance in KSC and Houston dataset, the T-SST-L gains 94.04% and 94.81% in OA index; the SSFTT gians 93.82% and 97.52% in OA index; the RSSAN gians 93.04% and 93.76% in OA index; and our proposed achieves 99.70% and 98.80%. This indicates that our method can still maintain stable classification performance in the large HSI dataset.

In addition, on IP, KSC and Houston, our method achieves more than a 2% improvement of OA, AA and kappa compared with HSI-BERT, SSAN and deep attention graph convolutional network (DAGCN), but on PU the improvement is not evident. The main reason for this is that the PU dataset has a sufficient number of examples on which to train the network, so all of the methods can obtain optimal results. This finding implies that our proposed framework can fully learn the feature distribution



(a) (b)

Fig. 13. Enlarged and visualized pixels classification results at the edge of some land-cover area as shown in (a) and (b) with different methods. (Left to right) are 2-D-CNN, HSI-BERT, DAGCN, SSAN, SST-M, and Ground Truth.

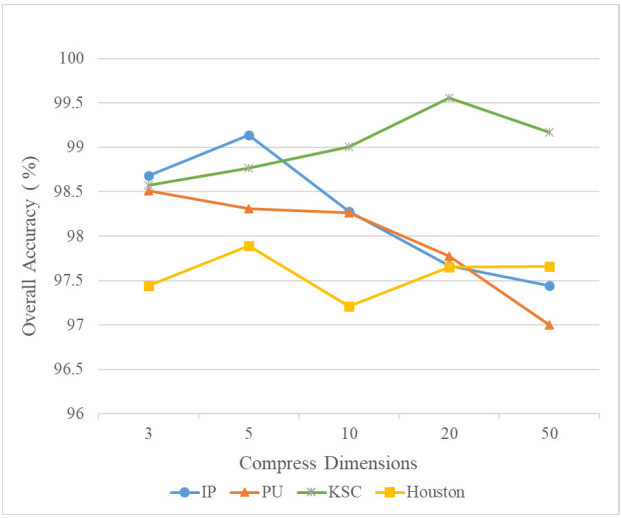


Fig. 14. OA results (%) of experiments with different numbers of convolution kernels in SFE net.

using only a small amount of the training data compared to other methods.

E. Discussion About SSA Model and SFE Net

The SFE net includes two convolution layers, which extract the spectral features and compress the HSI cube to different dimensions. We conducted comparative experiments to verify the impact of the dimensions compressed by the SFE net on the classification performance. Fig. 14 shows the overall accuracies of the HSI cube compressed to 3, 5, 10, 20, and 50 channels. The experimental results show that the best compression dimension of the SFE net is different among the four datasets. On the IP and Houston datasets, using five convolution kernels to extract features can obtain the best classification results, and on the KSC dataset, using 20 convolution kernels to extract features can obtain the best classification results. At the same time, the overall difference in classification performance is not very obvious under different spectral feature dimensions, which demonstrates that the SFE performance of SFE net does not depend on the number of channels of the final feature map, and the SFE net can effectively extract the spectral features of HSIs.

The SSA module provides attention weight to represent the spatial location importance. To prove that the spatial features

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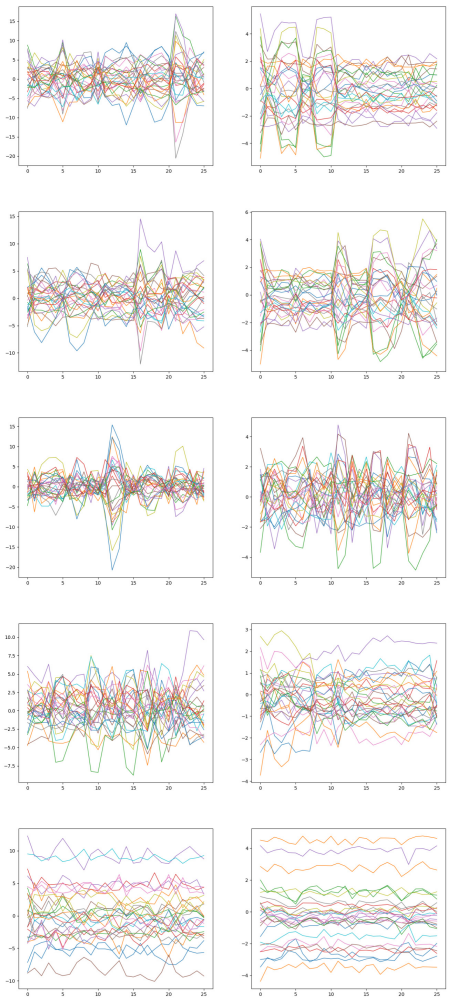


Fig. 15. Visualization results of features (Right) with SSA and (Left) without SSA.

are enhanced or weakened by adding weights, we visualize the features learned from the network. The left side of Fig. 15 shows the feature visualization result without adding the position weight, and the right side is the feature visualization result after adding the special position weight. The above figures in Fig. 15 show the visualization results of some randomly selected feature samples. In particular, the features extracted by the network have some changes after the attention weight is added. First, the value range for all of the vectors becomes smaller, and the feature curve converges to the center level. Second, for some feature curves, the amplitude and shape remain basically unchanged, but some feature curves obviously change, which shows that some features have indeed been enhanced, and some features have been weakened. In addition, the results of the ablation test also prove that our proposed patch attention model achieves the expected effect in the OA index.

F. Discussion of the MP Model

The MP model relabels all pixels in the HSI cubes for mask branch training, which undoubtedly requires many manual labeling efforts. In practice, labeling all pixels in HSIs requires considerable labor costs, which is undoubtedly undesirable. Therefore, we design another compromised mask generation method, in which we only label the category of the central pixel in the HSI cube and set all other positions to "0". The original method of complete labeling is masked as "mask_all", and the mask mode of labeling only the central pixel is marked as "mask_1". Experiments are also conducted to compare the classification accuracy of the network under the two mask labeling methods. The experimental results are shown in Table XII. We found that on IP and Houston datasets, the classification results of the network with mask_all are best. On the PU datasets, the classification result of the network with mask_1 is best. At the same time, the network with the mask branch shows better classification results on all datasets than the network without the mask branch. This shows that the mask branch we proposed effectively improves the classification ability of the network. We retain the two methods of mask generation, which can be flexibly used in actual conditions.

TABLE XII

CLASSIFICATION RESULTS OF THE PROPOSED METHOD WITH DIFFERENT MASK APPROACHES ON THE IP DATASET (%)

Datasets		mask_all	mask_1	No mask
IP	OA	99.08±0.31	98.73±0.15	98.68±0.13
	AA	99.01±0.33	97.12±0.27	95.03±0.54
	Kappa	98.59±0.36	98.49±0.39	98.50±1.15
PU	OA	99.61±0.28	99.69±0.03	98.51±1.79
	AA	99.23±0.65	99.23±0.08	97.57±2.63
	Kappa	99.48±0.37	99.60±0.04	98.01±2.41
KSC	OA	99.70±0.06	99.77±0.38	98.57±0.60
	AA	99.43±0.01	99.34±0.71	97.30±1.13
	Kappa	99.67±0.07	99.31±0.42	98.41±0.67
Houston	OA	98.80±0.13	97.95±0.73	97.44±1.01
	AA	98.51±0.16	98.03±0.92	97.24±0.24
	Kappa	98.71±0.15	98.46±0.34	97.24±1.09

In fact, for a HSI cube containing multiple pixel categories, the deep network cannot distinguish the target information and other interference information when extracting features. Only the loss between the predicted and real label and back propagating operates the model to recognize effective features. The role of the mask feedback branch is to provide another standard for the model. The Mask loss forces the network to pay attention to the center position and other positions with the same ground category as the central position, to improve the network's ability to distinguish the real category information. In addition, for the classes "Alfalfa", "Grass-pasture-mowed", and "oats" in the IP dataset, our method without MP branch obtains classification results of 98.05%, 100%, and 38.89%, respectively; and the results with MP is 99.27%, 100%, and 100%, respectively, This indicates that MP method is contribute to the model to accurately classify HSI pixel blocks with complex background.

In SST-M, the overall loss function consists of two parts: the classification prediction loss and the MP loss. Formula (14) represents the proportional distribution of the two losses in the overall loss. The experiments aim to verify the effect of α on the network. Fig. 16 shows the OA of the network when

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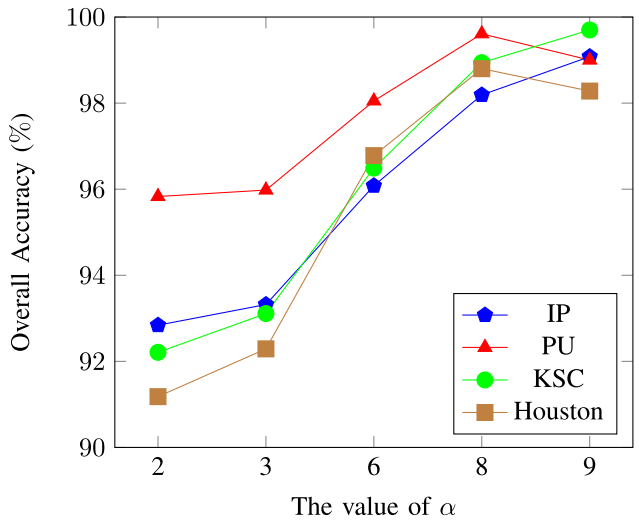


Fig. 16. OA result (%) of experiments with different values of alpha in formula (14).

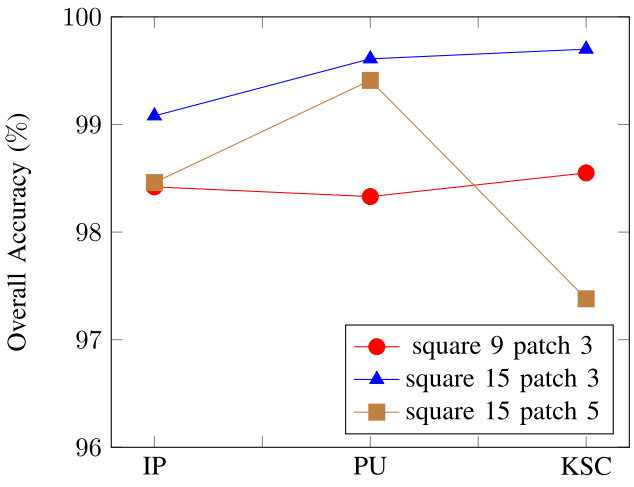


Fig. 17. Classification results for different square sizes and patch sizes.

the α is 2, 3, 6, 8, and 9 on the four datasets. It is obvious that the larger the value of α is, the higher the classification accuracy of the network; the best α for PU and Houston is 8, and the best α for IP and KSC is 9. The experimental results show that α should take a larger value. Because the purpose of the network is to classify accurately, the classification loss should occupy the main part of the overall loss.

To search for a perfect method of patch division, we set up a square region of 9×9 with a patch size of 3×3 and another square region with dimensions of 15×15 and a patch size of 5×5 to classify HSIs individually, as shown in Fig. 17. We list the OA of three datasets. The findings reveal that all of the methods with smaller square regions or larger patch sizes have a negative impact on the classification result. Hence, we argue that the best method of patch division should obey the rule that a small patch size and a large square region should be used. Meanwhile, the square size must be divided by the patch size so that we cannot divide HSI cubes arbitrarily; for example, HSI cubes with 7×7 and 11×11 sizes are not available.

In addition to the excellent performance of the SST-M, its time consumption also warrants attention. We have evaluated the time consumption of several methods in the training phase and the test phase, as shown in Table XIII. The SST-M is also much faster than some spatial-spectral-based methods. In the training phase, CNN is much faster than the others because CNN did not exploit the attention mechanism to use spatial information. In addition, the T-SST-L method requires extra time to extract spatial features for all of the spectral bands on three datasets, so it may not be a practical method. SSFTT is essentially a simple model integrate the CNN and transformer, only one SA layer is used in this model, so it has the same time complexity as the CNN. Benefitting from the relatively shallow architecture (only three layers in our manuscript) and the parallel implementation of each head in the MHSA mechanism, our network runs faster than other DL methods such as SSRN, which require a very deep architecture

TABLE XIII
COMPARISON TIME COMPLEXITY ON THREE DATASETS

	Indian Pines		PaviaU		KSC	
	Training(min)	Test(s)	Training(min)	Test(s)	Training(min)	Test(s)
CNN [12]	10.095	1.773	6.683	5.242	2.383	0.865
T-SST-L [45]	23.772	11.143	17.717	22.142	7.483	5.632
SSFTT [48]	12.688	6.125	9.398	4.565	2.387	1.262
SSRN [49]	25.247	12.380	18.749	37.101	6.332	5.572
RSSAN [50]	68.327	38.116	40.336	66.225	30.185	18.727
SST-M	10.969	3.432	8.386	10.110	9.411	2.998

to achieve satisfactory performance. Similarly, in the testing phase, our proposed method has the same excellent performance compared with other methods.

Furthermore, considering that there are often few training samples that can be used in practice, the proposed model is also evaluated with a small number of training samples. Table XIV lists the classification performance with different numbers of training data. These training data are composed of 20 randomly sampled samples per class, 50 samples per class, 1% samples per class, and 1% samples per class from the original data. Obviously, for any method, the smaller the amount of training data, the worse the classification accuracy will be. It is easy to see that the proposed method consistently performs better than the others. In addition, for any method, the smaller the amount of training data, the worse the classification accuracy.

In summary, the proposed method achieves the best performance for the OA, AA, and kappa coefficients of the compared methods. Abundant experiments verify that the SFE, SSA, and MP models are suitable for hyperspectral classification tasks. Experimental results show that the transformer structure exhibits a clear advantage with respect to extracted image features compared with 2-D CNNs and GCNs, and the spatial information and spectral channel features play a vital catalytic role for HSI classification. In addition, the classification maps are presented in Figs. 9–11. Our method shows little noise and few mistakes in classification, particularly for pixels located

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TABLE XIV
CLASSIFICATION RESULTS OF OUR PROPOSED METHOD AND OTHER LEADING METHODS ON THE KSC DATASET (%)

PU												
	1%			5%			20shots			50shots		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
CNN	95.85	**93.98**	94.5	98.37	97.95	97.01	90.86	89.84	87.83	95.26	94.93	93.70
HSI-Bert	95.22	91.15	93.65	98.15	96.47	97.22	90.88	91.84	89.69	97.44	97.24	96.6
SSRN	96.75	92.3	95.82	99.04	98.33	98.37	92.64	91.71	91.35	97.55	97.4	96.11
SST-M	**97.08**	92.76	**96.94**	**99.10**	**98.74**	**98.51**	**93.98**	**94.22**	**92.05**	**98.17**	**98.21**	**97.57**
KSC												
	1%			5%			20shots			50shots		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
CNN	84.12	74.029	82.3	91.38	85.07	90.41	88.71	84.13	87.38	99.19	98.789	99.09
HSI-Bert	-	-	-	92.07	88.02	91.16	93.58	91.23	92.83	99.54	99.391	99.48
SSRN	94.5	**94.86**	93.91	97.64	95.88	96.83	95.41	**95.47**	95.22	99.48	**99.54**	99.44
SST-M	**95.47**	91.32	**94.96**	**98.15**	**96.61**	**97.94**	**96.73**	94.6	**96.34**	**99.56**	98.95	**99.51**
Houston												
	1%			5%			20shots			50shots		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
CNN	75.83	74.47	73.85	93.01	92.89	92.44	84.21	86.29	82.94	90.21	91.67	89.42
HSI-Bert	66.26	62.34	63.47	96.04	95.11	95.71	83.35	85.61	82.04	95.23	94.69	94.31
SSRN	86.38	87.4	89.01	96.12	95.87	95.43	86.01	85.14	86.90	96.00	**96.23**	**96.5**
SST-M	**91.23**	**90.69**	**90.52**	**97.13**	**97.22**	**96.91**	**87.24**	**87.76**	86.16	**96.82**	**97.01**	96.45

in the edge region. It achieves more precise and fine-grained classification results.

V. CONCLUSION

In this article, we introduce a transformer encoder module combined with spatial-spectral features for HSI classification methods and achieve more precise classification effects. In addition, we fully utilize pixel label information in the HSI cube and propose a MP method to boost the fitting ability of our framework. Experiments on the IP, PU, KSC, and Houston datasets demonstrate that our framework shows remarkable performance compared with other leading methods. Our methods show that CNN-based modules can be replaced by transformer encoder modules. Meanwhile, because the multiheaded self-attention (MHSA) mechanism captures the long-range dependence in an arbitrary input cube, a deeper layer network can be abandoned without missing spatial relationships. In fact, our goal is to preserve the integrity of the transformer structure and only adjust the position embedding approach, leading to unexpected results, which serves as inspiration to further explore the transformer model optimization method and reduce the computational complexity.

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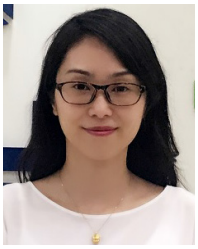
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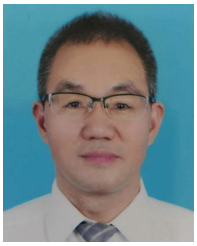
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